

Homework 04: Data frames and data wrangling

Please read the entire chapter on data transformation from [R for Data Science](#) before starting this homework.

This homework relies on the `nycflights13` package, which contains several data frames, including:

- `airlines`
- `airports`
- `flights`
- `planes`
- `weather`

Loading `nycflights13` puts these data frames on the search path.

Setup

Load packages (do this every time)

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.1      v tibble     3.3.1
v lubridate  1.9.4      v tidyr      1.3.2
v purrr      1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()      masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(nycflights13)
```

Note - if you don't have these installed you will need to first `install.packages('nycflights13')`

Question 1: Filtering

Make a plot of **air time vs distance** (air time on the y-axis, distance on the x-axis) for all flights that meet the following criteria:

- originate from LaGuardia airport ("LGA")
- departed on the 16th of the month
- have a flight distance of less than 2000 miles

Your code

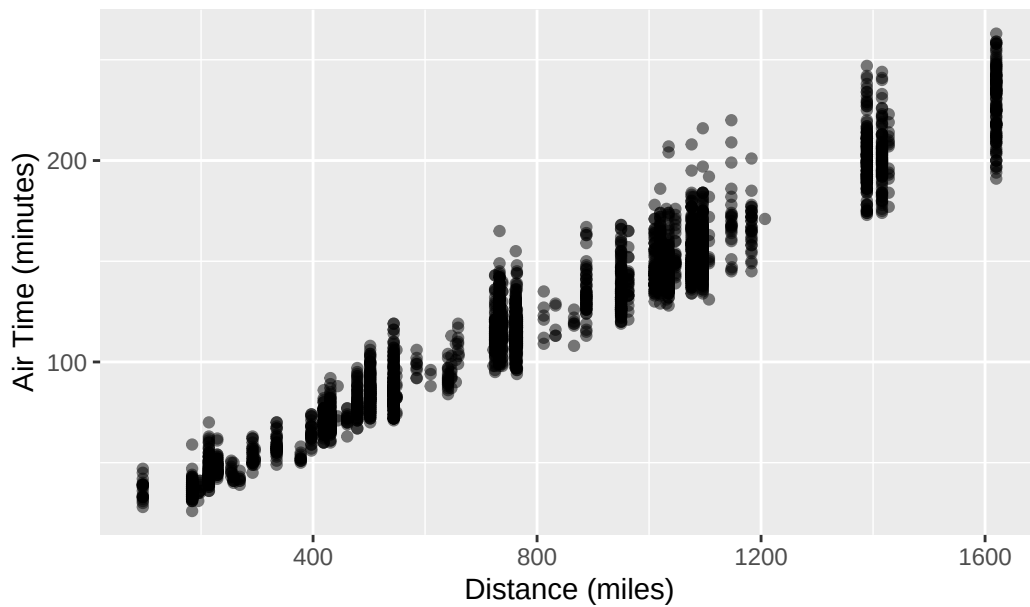
```
# Filter flights according to the criteria above,
# then make the required plot.

lga_flights_16 <- flights |>
  filter(origin == "LGA",
         day == 16,
         distance < 2000)

ggplot(lga_flights_16, aes(x = distance, y = air_time)) +
  geom_point(alpha = 0.5) +
  labs(title = "Air Time vs Distance (LGA, 16th of Month)",
       x = "Distance (miles)",
       y = "Air Time (minutes)")
```

Warning: Removed 75 rows containing missing values or values outside the scale range (``geom_point()``).

Air Time vs Distance (LGA, 16th of Month)



Brief written response

In 2–4 sentences, describe what you observe in the plot. (For example: Is the relationship roughly linear? Are there any clear outliers?)

Your answer here: There is a linear relationship with air time increasing as distance increases.

Question 2: Dealing with NAs

Make a data frame of all rows of `flights` that have values for *both* `arr_time` and `dep_time` (i.e., neither value is NA).

Your code

```
# Create a new data frame that removes rows where arr_time or dep_time is NA.

flights_no_na <- flights |>
  filter(!is.na(arr_time),
         !is.na(dep_time))
```

Filtering NAs (conceptual)

`ggplot()` will automatically remove NA values from a plot, but it emits a warning message about it. You *could* silence warnings using chunk options, but instead:

Brief written response

Explain (in words) how you could prevent those NA values from appearing in the plot in the first place.

Your answer here: To prevent NA values you have to filter them out before plotting. This removes rows with missing values so ggplot doesn't automatically drop them.

Question 3: Adding columns

Create a data frame of **average flight speeds**, based on `air_time` and `distance`.

Then make either:

- a histogram, **or**
- a density plot

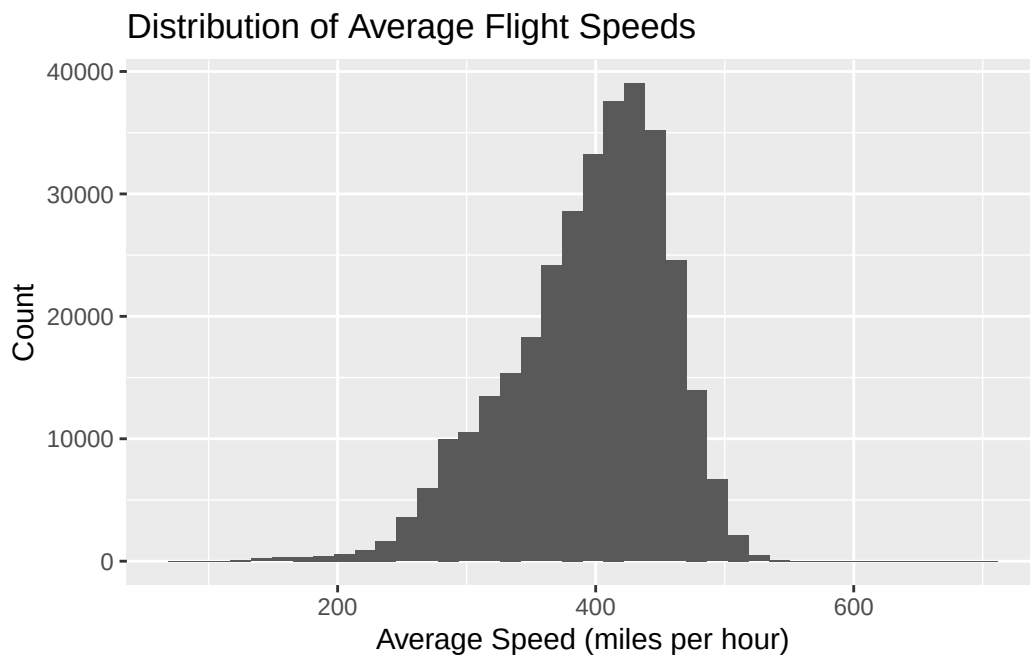
If you like, you may break the data out (e.g., by airline or another variable) in a way that you think makes sense.

Your code

```
# Create a new column for average speed.
# (Hint: think carefully about units - air_time is in minutes.)
# Then make a histogram or density plot.

flights_speed <- flights |>
  filter(!is.na(air_time), air_time > 0) |>
  mutate(speed = distance / (air_time / 60))

ggplot(flights_speed, aes(x = speed)) +
  geom_histogram(bins = 40) +
  labs(title = "Distribution of Average Flight Speeds",
       x = "Average Speed (miles per hour)",
       y = "Count")
```



Brief written response

Describe the main features of your speed distribution.

Your answer here: There is a bell shape, with most flights around 400 mph.

Rendering and submission (GitHub)

Canvas contains the course GitHub link and instructions for **forking** the course repository.

In this file, your job is to:

1. Complete the homework by adding code + written responses above.
2. Render this Quarto file to **HTML or PDF**.
3. Move the rendered file(s) to the /doc folder (note that, if you render to html you will also have to move your hmk_04_data_frames_files folder)
4. Add/commit both:
 - this edited .qmd file
 - the rendered output file(s) (HTML or PDF)
5. Push your changes to your fork on GitHub.
6. Open a Pull Request back to the course repository.

What to submit on Canvas

On Canvas, submit a link to your GitHub fork repository (not files).